WORKED SOLUTION

STEP III 2014, Question 2

Pure • Hyperbolic Integration

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q2

The key idea

Both integrands have $\cosh 2x$ underneath, and the double-angle formula can be written either in terms of $\cosh^2 x$ or in terms of $\sinh^2 x$. Choosing the version that matches the numerator turns each integral into a standard rational integral. For the final part, the trick is to go back to exponentials: with $u = \mathrm{e}^x$ both integrands become rational functions of u with $1 + u^4$ in the denominator, and a suitable combination of them gives exactly $1/(1+u^4)$.

Solution

Part (i)

We use $\cosh 2x = 2\cosh^2 x - 1$. With $u = \cosh x$ we have $\mathrm{d}u = \sinh x \,\mathrm{d}x$, so

$$\int \frac{\sinh x}{\cosh 2x}\,\mathrm{d}x = \int \frac{\mathrm{d}u}{2u^2 - 1}.$$

The denominator factorises as $(\sqrt{2}u - 1)(\sqrt{2}u + 1)$, and the two factors differ by 2, so

$$\frac{1}{2u^2-1} = \frac{1}{2}\left(\frac{1}{\sqrt{2}u - 1} - \frac{1}{\sqrt{2}u + 1}\right).$$

Each term integrates to a logarithm, with a factor $\frac{1}{\sqrt{2}}$ from the coefficient of u:

$$\begin{aligned}\int \frac{\mathrm{d}u}{2u^2-1} &= \frac{1}{2\sqrt{2}}\Big(\ln|\sqrt{2}u - 1| - \ln|\sqrt{2}u + 1|\Big) + C \\ &= \frac{1}{2\sqrt{2}} \ln\left|\frac{\sqrt{2}\cosh x - 1}{\sqrt{2}\cosh x + 1}\right| + C,\end{aligned}$$

which is the required result. (In fact $\cosh x \geq 1$ makes $\sqrt{2}\cosh x - 1 > 0$, so the modulus signs aren't needed for real x.)

Part (ii)

Now the numerator is $\cosh x$, which is the derivative of $\sinh x$, so the similar substitution is $u = \sinh x$, $\mathrm{d}u = \cosh x\,\mathrm{d}x$. Writing $\cosh 2x = 1 + 2\sinh^2 x$,

$$\int \frac{\cosh x}{\cosh 2x}\,\mathrm{d}x = \int \frac{\mathrm{d}u}{1+2u^2} = \frac{1}{\sqrt{2}}\arctan(\sqrt{2}\,u) + C$$
$$= \frac{1}{\sqrt{2}}\arctan(\sqrt{2}\sinh x) + C,$$

using $\int \mathrm{d}u/(1+a^2u^2) = \frac{1}{a}\arctan(au)$ with $a = \sqrt{2}$.

Part (iii)

At first sight there is no cosh anywhere in $\int_0^1 \frac{\mathrm{d}u}{1+u^4}$.

Ask yourself What does $\cosh x/\cosh 2x$ look like if we write everything in terms of $u = \mathrm{e}^x$?

Since $\cosh x = \frac{1}{2}(u + u^{-1})$ and $\cosh 2x = \frac{1}{2}(u^2 + u^{-2})$, multiplying top and bottom by u^2 gives

$$\frac{\cosh x}{\cosh 2x} = \frac{u+u^{-1}}{u^2+u^{-2}} = \frac{u(u^2+1)}{u^4+1}, \qquad \frac{\sinh x}{\cosh 2x} = \frac{u(u^2-1)}{u^4+1}.$$

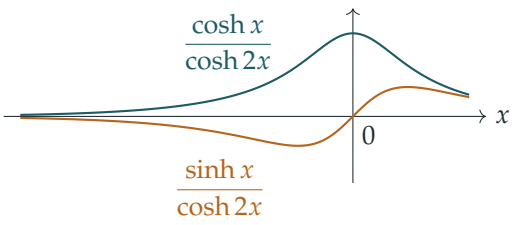
Also $\mathrm{d}x = \mathrm{d}u/u$, so the extra factor of u cancels:

$$\frac{\cosh x}{\cosh 2x}\,\mathrm{d}x = \frac{1+u^2}{1+u^4}\,\mathrm{d}u, \qquad \frac{\sinh x}{\cosh 2x}\,\mathrm{d}x = \frac{u^2-1}{1+u^4}\,\mathrm{d}u.$$

Subtracting the second from the first leaves $\dfrac{2}{1+u^4}\,\mathrm{d}u$. Finally, u running from 0 to 1 means $x = \ln u$ running from $-\infty$ to 0. Hence

$$\int_0^1 \frac{\mathrm{d}u}{1+u^4} = \frac{1}{2}\int_{-\infty}^0 \frac{\cosh x}{\cosh 2x}\,\mathrm{d}x - \frac{1}{2}\int_{-\infty}^0 \frac{\sinh x}{\cosh 2x}\,\mathrm{d}x.$$

Both integrals on the right converge, because each integrand is $O(\mathrm{e}^x)$ as $x \to -\infty$. (The left-hand side is an ordinary integral, so this is really just a change of variable, with the infinite end coming from $u \to 0$.)



The cosh integral. By part (ii), as $x \to -\infty$ we have $\sinh x \to -\infty$ and so $\arctan(\sqrt{2}\sinh x) \to -\frac{\pi}{2}$. Therefore

$$\int_{-\infty}^{0} \frac{\cosh x}{\cosh 2x}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\left(\arctan 0 - \left(-\tfrac{\pi}{2}\right)\right) = \frac{\pi}{2\sqrt{2}}.$$

The sinh integral. By part (i), as $x \to -\infty$ we have $\cosh x \to \infty$, so the fraction inside the logarithm tends to 1 and the logarithm to 0. At $x = 0$, $\cosh 0 = 1$, so

$$\int_{-\infty}^{0} \frac{\sinh x}{\cosh 2x}\,\mathrm{d}x = \frac{1}{2\sqrt{2}} \ln \frac{\sqrt{2}-1}{\sqrt{2}+1}.$$

Rationalising, $\dfrac{\sqrt{2}-1}{\sqrt{2}+1} = (\sqrt{2}-1)^2$, and $\sqrt{2}-1 = \dfrac{1}{\sqrt{2}+1}$, so

$$\int_{-\infty}^{0} \frac{\sinh x}{\cosh 2x}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\ln(\sqrt{2}-1) = -\frac{1}{\sqrt{2}}\ln(\sqrt{2}+1).$$

This is negative, as it should be: $\sinh x < 0$ for $x < 0$ (see the amber curve).

Putting it together.

$$\int_0^1 \frac{\mathrm{d}u}{1+u^4} = \frac{1}{2}\left(\frac{\pi}{2\sqrt{2}} + \frac{1}{\sqrt{2}}\ln(\sqrt{2}+1)\right) = \frac{\pi + 2\ln(\sqrt{2}+1)}{4\sqrt{2}},$$

as required.

Checks

- Differentiate the answers to (i) and (ii) and recover the integrands; for (ii), $\frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{\sqrt{2}}\arctan(\sqrt{2}\sinh x) = \dfrac{\cosh x}{1+2\sinh^2 x}$.
- The final value is ≈ 0.86697, which agrees with a Simpson's-rule evaluation of $\int_0^1 \mathrm{d}u/(1+u^4)$. It sits sensibly between $\frac{\pi}{4} \approx 0.785$ (the integral of the smaller function $1/(1+u^2)$) and 1.
- Separately, the two infinite integrals were confirmed numerically: $\pi/(2\sqrt{2}) \approx 1.1107$ and $-\ln(1+\sqrt{2})/\sqrt{2} \approx -0.6232$.

Common slips

- Using $\cosh 2x = 2\cosh^2 x - 1$ in part (ii) as well: with $u = \sinh x$ you need the $\sinh^2$ form, $1 + 2\sinh^2 x$.
- Forgetting that $\mathrm{d}x = \mathrm{d}u/u$ when $u = \mathrm{e}^x$; it is exactly this factor that cancels the stray u in the numerators.
- Getting the limits wrong: $u \in [0,1]$ corresponds to $x \in (-\infty, 0]$, not $[0,1]$.

Beyond STEP

The same integral can be done directly with partial fractions, since $1+u^4 = (u^2+\sqrt{2}u+1)(u^2-\sqrt{2}u+1)$; the combination $\frac{1+u^2}{1+u^4}$ and $\frac{1-u^2}{1+u^4}$ is a neat shortcut that avoids most of that algebra. At university, integrals like $\int_0^\infty \mathrm{d}u/(1+u^4) = \pi/(2\sqrt{2})$ are done in one line by contour integration and the residue theorem, and improper integrals are handled with full rigour in analysis. Needham's *Visual Complex Analysis* is a beautiful place to see the complex-analytic side.

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